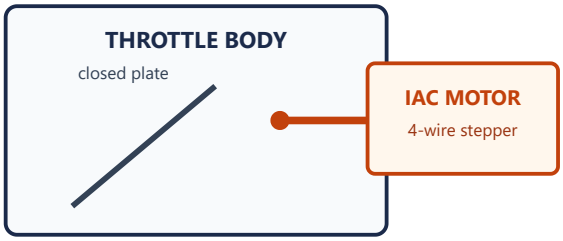
P0505 AND IDLE-AIR CONTROL

IAC operation and P0505

The idle air control motor is a stepper on the throttle body. The throttle plate is closed at idle. Air the engine needs at idle goes through a bypass passage. The PCM moves a pintle in steps and changes how much air gets through.

There is no pintle-position sensor. The PCM counts the steps it has commanded from a reference. It does not "see" where the tip actually is. That is why a pintle moved by hand is a problem even when the coils still measure correctly.

Cold idle is supposed to be higher. A/C, the cooling fan, and an automatic transmission in Drive are loads. The PCM opens the bypass to hold RPM. Judge the motor only when the engine is warm and you have written down the loads.



DO NOT PUSH OR PULL THE PINTLE

Not by hand. Not with a pick. Not "to seat it."

The pintle on this truck was moved by hand STATED

Treat that motor as damaged. Pushing or pulling the pintle can strip the internal screw, bend the tip, or leave the bypass parked where the PCM's step count is wrong. The result can be a high idle, a stall, a hunt, or a circuit fault if the mechanism is jammed. Those are POSSIBLE outcomes, not a list of things already measured. An ohms test cannot see a stripped screw. Replace it with a **new** motor. Do not install a used one. Do not touch the pintle on the new one, in the box or on the truck.

What P0505 is, and is not

CONFIRMED Generic title: idle control system malfunction. On this vehicle family the code is often an **electrical** monitor: a short to ground or to battery on an IAC driver circuit while the motor is commanded. LIKELY Read the scan tool's manufacturer description and save the freeze frame. A high idle alone may set a different monitor, often described as P1294, target idle not reached. TEST REQUIRED P0505 does not, by itself, prove a vacuum leak. Check the circuit and the leaks. Both can be present.

P0505 · VACUUM LEAK VERSUS THE IAC

High idle: leak, or the IAC?

Clue	Points more toward a vacuum leak or a plate that is not shut	Points more toward the IAC
RPM	Stays high even when a scan tool shows the IAC driven to its closed, low-air stop. The exact step count is calibration data. Do not invent a number. TEST REQUIRED	RPM follows IAC command, or the idle is wrong and the pintle was pushed, or the coils measure open or shorted.
Unplugging the IAC	If the pintle is already closed and RPM stays very high, extra air is coming from somewhere else.	If unplugging a motor that was held open drops the idle sharply, that motor was the air path. Freezing a bad position is not a fix.
Sound and smoke	Hiss at an elbow, the booster hose, the intake gasket, or smoke exiting one of those joints.	Smoke only at the IAC base means the O-ring or the mounting face, which is still an air leak at the IAC.
Fuel trims	On this MAP engine, trims may stay near normal because the PCM sees the higher MAP and adds fuel. Trims that go very positive only at idle still deserve a leak and exhaust-leak check.	Trims do not test the stepper windings. Use ohms and the step command for that.
History on this truck	Brittle elbows and a cracked CCV hose are common on this engine. LIKELY pattern. Confirm with smoke.	The pintle was moved. That history is enough to replace the motor even if ohms look fine.

Do the leak check before the idle relearn. A relearn on a leaking engine teaches the PCM the leak. Guide 04. Fix what smoke and the visual inspection find. Then install the new IAC. Then relearn.

IAC resistance TEST REQUIRED

Two coils, four pins. Measure one coil across one pair and the other coil across the other pair. On a straight 4-pin row the usual pairs are the two outer pins and the two inner pins. Several 4.0L procedures cluster near **48–60 Ω per coil**. LIKELY as a field figure, not a 2004 FSM limit in hand. **Screening window for this job: about 40–80 Ω per coil.** The two coils should not read a few ohms to each other. Neither coil should read to the motor case.

Reading	Call
Both coils inside 40–80 Ω, coils isolated from each other	Windings are electrically plausible. If the pintle was pushed, replace the motor anyway. The screw can be ruined with good ohms.
OL on either coil	Open winding. Replace. Do not try to “free” it by pushing the pintle.
Near 0 Ω on a coil, or a few ohms between the two coils	Shorted winding. Replace. Inspect the harness for a short that may have caused it, or that it may have caused.

Key-on voltage at the IAC connector is a PCM driver output, and it changes as the drivers step. Two wires may sit near battery voltage and two near ground, and then swap. A single frozen reading is not a full test. Use it only as a “is anything there” check, then go back to ohms on the motor and a wiggle test of the connector. Wire colors are not listed here on purpose.

P0505 · THROTTLE BODY AND INSTALLATION

Clean the bore. Install a new IAC.

New motor only. Not a junkyard motor, not a motor whose pintle was pushed back in the box, not the original after the pintle was moved. Catalog candidate: Mopar **4874373AB**, listed for Grand Cherokee idle-speed control, 1998–2004. CATALOG LISTED Verify by VIN. This PCM has a reported intolerance for poor aftermarket IACs. POSSIBLE Prefer the new Mopar motor.

Throttle-body inspection and cleaning

1. Key off. Remove the air duct at the throttle body so you can see the plate. Disconnect the IAC connector before any spray.
2. Look for carbon on the plate edge and a plate that does not sit closed. Look at the cable and the cam. The plate must return fully when you open it by the cam and let go. A cable that holds the plate open makes a high idle no IAC can correct.
3. Hold the plate open with the cam. Spray throttle-body cleaner on a shop towel and on the bore. Do not flood the TPS or the IAC connector. Do not pry the plate with a screwdriver.
4. With the IAC removed, clean the bypass passage in the throttle body. Do not spray out a new IAC and do not push its pintle to clean behind the tip. The old motor is not being saved if the pintle was forced.
5. Wipe the bore dry enough that solvent will not be ingested on the first start. Confirm the plate closes and the cable returns.

Installation

1. Negative battery cable off and secured, so the PCM is not stepping a motor you are handling. This also starts the clock for the relearn on the next page.
2. Two screws hold the motor. They are commonly a T20 Torx. Reported torque is **7 N·m (60 in-lb)**. LIKELY Do not overtighten into the aluminum.
3. The O-ring must be present, clean, and not twisted. A missing ring is a vacuum leak at the IAC base. Do not oil it. A dry ring seats.
4. Set the motor straight into the bore. If it does not sit flush, the ring is pinched or the pintle is against the passage. **Do not push the pintle to make it fit.** Remove it and look. The new pintle stays where the factory left it.
5. Snug the screws to the torque above. Connect the plug until the latch clicks. Reinstall the air duct. The duct clamps and the CCV hose at the duct are themselves leak points. Seat them.

After the battery is reconnected, turn the key on and wait about 10 seconds before you crank. Do not touch the accelerator. That key-on pause is the practical way to let this PCM move the pintle before the first start. It is shop practice for this cable-throttle engine, not a quoted FSM script. LIKELY

P0505 · PCM RESET AND IDLE RELEARN

PCM reset and idle relearn

Do this only after leaks are fixed, the throttle plate closes, and the new IAC is installed with the pintle untouched. Resetting first teaches the computer the fault. A scan tool that can clear Chrysler adaptive fuel and idle memory is the cleaner method if you have one. The battery method below is the practical standby.

This is not an electronic-throttle relearn. There is no accelerator-pedal procedure to perform. Foot off the pedal the entire time, except the brake when you select Drive.

1. Negative battery cable disconnected and tied so it cannot touch the post. Wait **15 minutes** minimum. Some radios ask for an anti-theft code afterward. Have that code before you start if this truck uses one.
2. Reconnect the battery. Key on, engine off, about 10 seconds. Do not touch the pedal. Then start.
3. Park, parking brake set, wheels chocked. A/C off, rear defrost off, blower off. Let it idle with no throttle input until the engine is fully warm. That can take 10 to 15 minutes. Do not blip the throttle to “help it.”
4. When warm, idle about two minutes with the A/C on, then turn the A/C off and let it settle.
5. If the brakes hold and the shifter is safe: foot on the brake, idle in Drive about two minutes with loads off, then briefly with the A/C on. Return to Park. Skip this step if the vehicle is not safe to put in gear.
6. Shut off. Restart. Read warm idle in Park, loads off. Screening band **700–800 rpm**. Factory software reference about **744 rpm** fully warm on the TSB calibration. Record RPM, coolant temperature, and gear on Guide 01’s test sheet.

How to read the result

Warm idle after the relearn	Meaning
About 700–800 rpm in Park, loads off, and it recovers after the A/C comes on	Idle control is plausible. **SCREEN MET** Rescan. A pending code still needs the electrical tests.
Still above about 1,000 rpm, no load	Extra air or a motor the PCM cannot close. Go back to smoke, the plate, the cable, and the IAC O-ring. Do not keep resetting.
Hunts more than about 100 rpm, warm, no load	A remaining leak, a dirty plate edge, a poor IAC, or a vacuum hose collapsing. **TEST REQUIRED**
Stalls as it warms, or only with the A/C on	IAC not opening under load, a connector not latched, or an air leak the PCM cannot trim out. Check the plug and command with a scan tool before you buy a second motor.
Idle is normal and P0505 returns immediately	Treat it as the circuit fault: wiring, connector tension, or a PCM driver. A good idle does not clear a shorted driver wire.

If the manifold starts to glow during this idle, shut it off. Guide 05. Do not finish the relearn on an engine that is overheating the exhaust.